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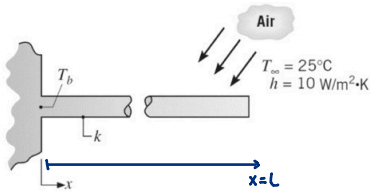
Heat Transfer HW #5

1. (60 pt) A cylindrical pin fin with a radius of $r = 1$ mm and length of $L = 5$ cm is exposed to airflow with a temperature of $T_\infty = 25$ °C and a heat transfer coefficient of $h = 10$ W/m²·K. Thermal conductivity of the pin fin is $k = 200$ W/m·K. The tip of the pin fin is thermally insulated. Use the fin problem solving workflow discussed in the lecture to solve the following questions.



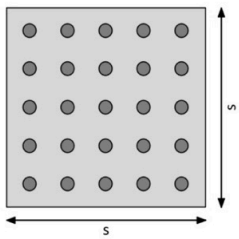
(a) If the fin base temperature is $T_b = 125$ °C, please calculate the fin temperature at $x = L/2$ and $x = L$.

(b) Calculate the fin effectiveness ε_f and the fin efficiency η_f based on their definition and results provided by Table 3.4 of the textbook.

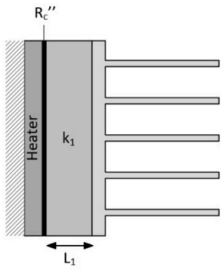
(c) Calculate the fin efficiency η_f based on results provided by Table 3.5. Compare the results of (b) and (c) and discuss if Table 3.5 provided a good approximation to the fin efficiency.

(d) Calculate the thermal resistance of this pin fin R_f.

(e) The pin fins then form a 5×5 array mounted on a base plate with a side length of $s = 3$ cm, as shown in the figure on the right-hand side. Calculate the overall efficiency of the fin array η_o and the thermal resistance of the fin array R_o. Calculate how much heat is dissipated through the unfined surface and how much is through the fins.



(f) The heat sink consisting of this fin array is used to improve the heat dissipation from a thin film heater, as shown in the figure on the right-hand side. The heat sink is mounted on a slab with thickness $L_1 = 2$ cm and thermal conductivity $k_1 = 10$ W/m·K. There is a thermal adhesive between the thin film heater and the slab, resulting in a contact thermal resistance $R_c = 10^{-4}$ m²·K/W. Thermal insulation was applied to the backside of the heater. The heater generates a uniform heat flux $q'' = 1000$ W/m², which was dissipated through the heat sink. The temperature across the thin film heater is uniform. Thermal resistance for the base of the heat sink can be neglected. There is no contact thermal resistance between the base of the heat sink and the slab. Please draw the thermal resistance network of this system and calculate the temperature of the thin film heater T_h. Compared with the condition without the heat sink, *i.e.*, directly exposing the heater and the slab to ambient air, how much temperature reduction of the heater is achieved?



GIVEN:

$r = 0.001$ m
$L = 0.05$ m
$T_\infty = 25°C$
$h = 10$ W/m²K
$k = 200$ W/mK
$T_b = 125°C$

ASSUMPTIONS:

- 2D heat transfer
- steady state
- no h.t. due to ΔPE or ΔKE
- constant air & material properties
- thermally insulated tip

part f:
- uniform temp. accross film heater
- neglect thermal resistance for the base
- no contact resistance between base & slab

SCHEMATIC:

in problem statement

FIND:

a) T_f @ $x = \frac{1}{2}L$ & $x = L$
b) fin effectiveness ε_f and fin efficiency η_f from table 3.4
c) fin efficiency η_f, compare a & b
d) R_f
e) η_o, R_o, dissipated heat through surface & through fins
f) draw thermal resistance network
calculate T_h, temp. reduction

ANALSIS:

a) for thermally insulated tips (adiabatic):

$$\left.\frac{d\theta}{dx}\right|_{x=L} = 0, \quad \frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh mL}, \quad q_f = M\tanh(mL), \quad \eta_f = \frac{\tanh(mL)}{mL}, \quad R_f = \frac{1}{\eta_o h A_f}$$

$P = 2\pi r = 2\pi(0.001\text{ m}) = 0.0063$ m

$A_c = \pi r^2 = \pi(0.001\text{ m})^2 = 3.14\cdot10^{-6}$ m²

$A_f = P\cdot L = 0.0063\text{m}\cdot 0.05\text{m} = 3.15\cdot10^{-4}$ m²

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(10\text{ W/m}^2\text{K})(0.0063\text{m})}{(200\text{ W/mK})(3.14\cdot10^{-6}\text{m}^2)}} = 10.02\text{ m}^{-1}$$

$$M = \sqrt{hPkA_c}\,\theta_b = \sqrt{(10\text{ W/m}^2\text{K})(0.0063\text{m})(200\text{ W/mK})(3.14\cdot10^{-6}\text{m}^2)}\,(125°C - 25°C) = 0.629\text{ W}$$

↳ $\theta_b = \theta(0) = T_b - T_\infty = 125°C - 25°C = 100°K$

↳ $\Delta°C = \Delta°K$

$$\theta(x=\tfrac{1}{2}L) = \frac{\cosh m(L-x)}{\cosh mL}\,\theta_b = \frac{\cosh\left[10.02\,m^{-1}(0.05m - \tfrac{1}{2}0.05m)\right]}{\cosh(10.02\,m^{-1}\cdot 0.05m)}\cdot 100°C = 91.44°C$$

$$\theta = T - T_\infty \Rightarrow T = \theta + T_\infty = 91.44°C + 25°C = 116.44°C \quad @\ x = \tfrac{1}{2}L$$

$$\theta(x=L) = \frac{\cosh m(L-x)}{\cosh mL}\,\theta_b = \frac{\cosh\left[10.02\,m^{-1}\cancel{(0.05m - 0.05m)}^{\,0}\right]}{\cosh(10.02\,m^{-1}\cdot 0.05m)}\cdot 100°C = 88.64°C$$

$$\Rightarrow T = \theta + T_\infty = 88.64°C + 25°C = 113.64°C \quad @\ x = L$$

b) $$\varepsilon_f = \frac{\text{fin heat rate}}{\text{heat rate of base w/o fin}} = \frac{q_f}{hA_c\theta_b} = \frac{0.291\,W}{(10\,W/m^2K)(3.14\cdot 10^{-6}m^2)(100°K)} = 91.66 = \varepsilon_f$$

$$\hookrightarrow q_f = M\tanh(mL) = (0.629\,W)\tanh(10.02\,m^{-1}\cdot 0.05m) = 0.291\,W$$

$$\eta_f = \frac{q_f}{q_{max}} = \frac{q_f}{hA_f\theta_b} = \frac{0.291\,W}{(10\,W/m^2K)(3.15\cdot 10^{-4}m^2)(100°K)} = 0.924 = \eta_f$$

c) TABLE 3.5:

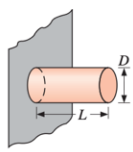
Pin Fins

Rectangular[b]

$A_f = \pi D L_c$

$L_c = L + (D/4)$

$V = (\pi D^2/4)L$



$$\eta_f = \frac{\tanh mL_c}{mL_c}$$

$m = \text{same} = 10.02\,m^{-1}$

$L_c = L + \tfrac{1}{4}D = 0.05m + 0.25(2\cdot 0.001m) = 0.0505m$

$$\eta_f = \frac{\tanh(mL_c)}{mL_c} = \frac{\tanh(10.02\,m^{-1}\cdot 0.0505m)}{(10.02\,m^{-1}\cdot 0.0505m)} = 0.922 = \eta_f$$

The fin efficiency using table 3.5 is a good approximation.
The calculated efficiencies from b & c are very close.

↳ the assumption works when fin is long & A_c is small relative to the length

d) $$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{(0.924)(10\,W/m^2K)(3.15\cdot 10^{-4}m^2)} = 343.57\ K/W$$

e) $s = 0.03m$

$$A_{tot} = NA_f + \underbrace{A_{uf}}_{s^2 - NA_c} = 25(3.15\cdot 10^{-4}m^2) + (0.03m^2 - 25(3.14\cdot 10^{-6}m^2) = 0.00868\,m^2$$

$$q_{tot} = Nq_f A_f h\theta_b + hA_{uf}\theta_b$$

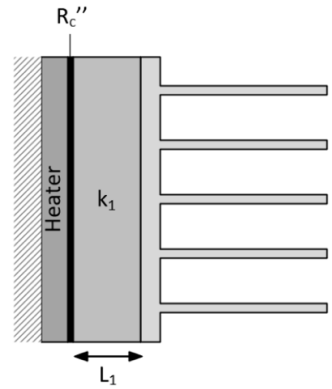
$$\eta_o = \frac{q_{tot}}{q_{max}} = \frac{Nq_f A_f \cancel{h}\cancel{\theta_b} + \cancel{h}A_{uf}\cancel{\theta_b}}{\cancel{h}A_{tot}\cancel{\theta_b}} = \frac{Nq_f A_f + A_{uf}}{A_{tot}}$$

$$= 1 - \frac{NA_f}{A_{tot}}(1-\eta_f)$$

$$= 1 - \frac{25(3.15\cdot 10^{-4}m^2)}{0.00868\,m^2}(1-0.924) = 0.931 = \eta_o$$

$$R_o = R_o = \frac{1}{\eta_o h A_{tot}} = \frac{1}{(0.931)(10\,W/m^2K)(0.00868m^2)} = 12.37\ K/W$$

f)

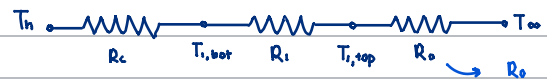


$L_1 = 0.02\,m$ $\qquad A = s^2 = 9 \cdot 10^{-4}\,m^2$

$k_1 = 10\,W/mK$

$R_c'' = 10^{-4}\,m^2\,K/W$

$q'' = 1000\,W/m^2$



R_o already accounts for all surface resistance from fin array and base plate to the air (only convection)

$$q = q'' \cdot A = 1000\,W/m^2 \cdot 9 \cdot 10^{-4}\,m^2 = 0.9\,W$$

$$R_c = R_c''/A = 10^{-4}\,m^2K/W \div 9 \cdot 10^{-4}\,m^2 = 0.111\,K/W$$

$$R_{tot} = R_c + R_1 + R_o = 0.111\,K/W + \frac{L_1}{k_1 A} + 12.37\,K/W$$

$$= 0.111\,K/W + \underbrace{\frac{(0.02\,m)}{(10\,W/mK)(9 \cdot 10^{-4}\,m^2)}}_{R_1 = 2.22\,K/W} + 12.37\,K/W$$

$$= 14.71\,K/W$$

$$\Delta T = q \cdot R_{tot} = 0.9\,W \cdot 14.71\,K/W = 13.24\,C \text{ or } K$$

$$T_h = T_\infty + \Delta T = 25°C + 13.24°C = 38.24°C$$

without heat sink: $\quad R_{tot} = R_c + R_1 + \frac{1}{hA} = 0.111\,K/W + 2.22\,K/W + \frac{1}{(10\,W/m^2 \cdot K)(9 \cdot 10^{-4}\,m^2)}$

$$= 113.44\,K/W$$

$$\Delta T = 0.9\,W \cdot 113.44\,K/W = 102.1°C$$

$$T_h = T_\infty + \Delta T = 25°C + 102.1°C = 127.1°C$$

temp reduction $= 127.1°C - 38.24°C = 88.86°C$ = temp reduction

2. (30 pt) A long wire of diameter $D = 1$ mm is submerged into an oil bath of temperature $T_\infty = 20$ °C. Density ρ, thermal conductivity k, and heat capacity c of the wire are 4000 kg/m^3, 400 W/m·K, and 500 J/kg·K, respectively. The wire has an electrical resistance per unit length of $R_e' = 0.01$ Ω/m. The oil bath can dissipate heat with a heat transfer coefficient of $h = 500$ W/m^2·K. At the initial moment $t = 0$ s, a current of $I = 100$ A flows through the wire and produces Joule heating.

(a) Calculate the steady-state temperature of the wire.

(b) Apply the lumped capacitance method to derive an expression for the wire temperature as a function of time. How long does it take for the wire to reach a temperature that is within 5 °C of the steady-state temperature?

GIVEN:

$D = 0.001\,m$

$T_\infty = 20°C$

$\rho = 4000\,kg/m^3$

$k = 400\,W/m\cdot K$

$c = 500\,J/kg\cdot K$

$R_e' = 0.01\,\Omega/m$

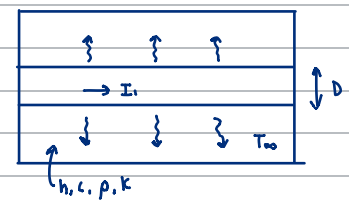
$I(t=0) = 100\,A$

$h = 500\,W/m^2K$

ASSUMPTIONS:

- 2D heat transfer
- steady state
- no h.t. due to ΔPE or ΔKE
- constant air & material properties

SCHEMATIC:



FIND:

a) steady state temp. T_s

b) $T(t)$, time for $T = T_s \pm 5°C$

ANALYSIS:

a) $q = I^2 R_e' = (100A)^2 (0.01\,\Omega/m) = 100\ W/m$

$q = h\pi D(T - T_\infty) \Rightarrow T_s = \frac{q}{h\pi D} + T_\infty = \frac{(100\ W/m)}{(500\,W/m^2K)\pi(0.001m)} + 20°C = 83.7°C$

b) $L_c = \frac{V}{A_s} = \frac{\pi r^2 L}{2\pi r L} = \frac{1}{2}r = \frac{0.001\,m}{4} = 0.00025m$

$Bi = \frac{hL_c}{k} = \frac{500 \cdot 0.00025m}{400} = 0.0003 < 0.1$ ✓

$\dot{E}_g - q_{conv} = \rho V_c \frac{dT}{dt}$

$\tau = \frac{\rho V c}{h A_s} = \frac{\rho c D}{4h} = \frac{(4000\,kg/m^3)(500\,J/kg\cdot K)(0.001\,m)}{4(500\,W/m^2\cdot K)} = 1.0s$

$\frac{d\theta}{dt} = \frac{dT}{dt} \Rightarrow \frac{\theta}{\theta_i} = e^{(-t/\tau)} = \frac{T - T_s}{T_i - T_s}$

$\Rightarrow T = T_s + (T_i - T_s)e^{-t/\tau} = 78.7°C \leftarrow 5\%\ T_s$

$\Rightarrow t = -\ln\left(\frac{5}{63.7}\right) = 2.55s$